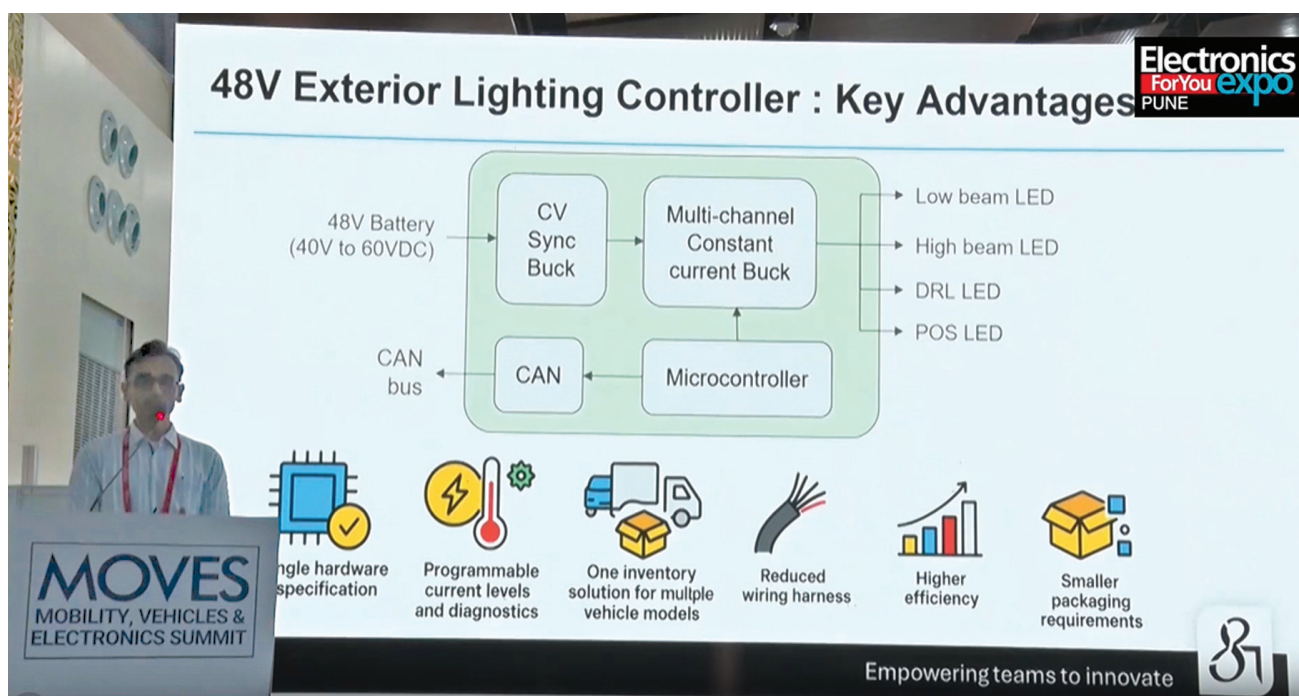


48V ARCHITECTURES

As The Next Power Step IN AUTOMOTIVE DESIGN

A new power paradigm is taking shape in automotive design, unlocking higher efficiency and simpler architectures while raising a fundamental question: Is the 12V system still relevant?



Mahendra Patel, Electronic Design Consultant, AIGHTECH Services Pvt Ltd, India during presentation at EFY Expo Pune

It is genuinely exciting to witness the pace at which India's semiconductor and electronics ecosystem is evolving. Across recent exhibitions, one trend has stood out clearly: the shrinking turnaround time from prototype to production. This signals more than faster execution; it reflects a maturing ecosystem that is learning from neighbouring markets, particularly China, whose remarkable pace offers lessons that can be thoughtfully adapted to the Indian context.

Rethinking the 12V legacy

As the ecosystem evolves rapidly, it is equally important for our hardware to keep pace with these shifts. The automotive sector is a good example. Vehicles once ran on 6V systems, then moved to 12V lead-acid batteries as electrical loads increased, and this eventually became the standard. Today, there is a clear shift towards 48V systems. Modern two-wheelers and three-wheelers are already moving to

48V, and even mild hybrid cars are adopting 48V batteries.

But are we really transitioning everything to 48V? The answer is no. Most systems still carry both 12V and 48V. That is where I start to question this approach. With the semiconductor solutions available today, do we really need both battery systems in the same vehicle? To me, this also means we need to challenge the long-held assumption that 12V must always be retained. That is no longer necessarily true.